

Ökobaudat EPD Matching — Generic Roof (200 mm)

Method: LLM layer decomposition + Ökobaudat REST API · No API key required · Database: OEKOBAUDAT 2021-II

Two approaches submitted for Q2. This document presents the *simple approach*: Claude (LLM) decomposes the roof into standard layers, then a minimal Python script queries the public Ökobaudat REST API to find the best matching EPD per layer — no Anthropic API key required. A separate document covers the full automated pipeline with standards verification.

SECTION 01

Input Element

ELEMENT DEFINITION AS GIVEN

name	Generic Roof
elementType	Roof
thickness	200 mm total
Roof type assumed	Flat roof (Flachdach) – most common generic case

SECTION 02

Method — Three Steps

01 Layer decomposition via LLM (Claude)

Claude is asked to decompose a generic flat roof at 200 mm total thickness into standard material layers, ordered inside → outside, per DIN 18531, DIN 4108-2, DIN 4108-3, DIN 1045, and GEG 2024. Minimum insulation thickness is respected first; remaining thickness is distributed across structural and finishing layers.

02 Ökobaudat REST API search per layer

A short Python script queries the public Ökobaudat API endpoint by German keyword for each layer. If the primary keyword returns no results, a fallback keyword is tried. The API returns up to 5 candidate EPD records per query — no authentication required.

03 Best match selection

The best EPD is selected by matching material type, density class, and intended application against the layer specification. A confidence level (HIGH / MEDIUM) and one-sentence reasoning are stated for each match.

SECTION 03

Assumed Layer Build-up

#	LAYER (INSIDE → OUTSIDE)	THICKNESS	STANDARD	SEARCH KEYWORD
1	Structural slab (Stahlbeton)	120 mm	DIN 1045	Beton
2	Vapour barrier (Dampfsperre)	5 mm	DIN 4108-3	Bitumenbahn
3	Thermal insulation (Wärmedämmung)	60 mm	DIN 4108-2 / GEG 2024	Mineralwolle
4	Waterproofing membrane (Dachdichtung)	5 mm	DIN 18531	Abdichtung
5	Gravel ballast (Kiesschüttung)	10 mm	DIN 18531	Kies
Total				200 mm ✓

60 mm mineral wool at $\lambda = 0.035 \text{ W/mK}$ gives $R = 1.71 \text{ m}^2\text{K/W}$ — a reasonable minimum for a simplified flat roof. A GEG-compliant design would typically use 120–160 mm; 60 mm is adopted here to fit the 200 mm total.

SECTION 04

EPD Match Results

LAYER	MM	BEST EPD MATCH	UUID	CONF.
Structural slab	120	Beton C20/25	bb6970aa-9d71-430f-8b81-930b9fe467c8	MED
Vapour barrier	5	Bitumenbahnen G 200 S4	bdf0a7e9-3fae-4780-b5ca-3d686c6d46d0	HIGH
Thermal insulation	60	Mineralwolle (Boden-Dämmung)	58b1ba79-459b-4d58-bc39-91a8f98e71ba	HIGH

LAYER	MM	BEST EPD MATCH	UUID	CONF.
Waterproofing membrane	5	PE-HD mit PP-Vlies zur Abdichtung	2173d540-fc7c-4b77-af4f-9a53cdbfb834	MED
Gravel ballast	10	Kies (Korngröße 2/32)	ed8da770-fc78-40cd-acb1-40f062330f9c	HIGH
Total	200			

Datastock: OEKOBAUDAT 2021-II UUID: cd2bda71-760b-4fcc-8a0b-3877c10000a8 · All EPDs retrieved via public REST API, no authentication required.

SECTION 05

Match Reasoning per Layer

LAYER	CONFIDENCE	REASONING
Structural slab	MEDIUM	Beton C20/25 is the closest generic concrete EPD available. Reinforcement steel is not separately declared in this datastock — in a full LCA, rebar would be added as an additional layer at approx. 100 kg/m³.
Vapour barrier	HIGH	G 200 S4 is a standard polymer-modified bitumen sheet (SBS) — the most common vapour barrier product for flat roofs in Germany per DIN 4108-3. Direct material and application match.
Thermal insulation	HIGH	Mineralwolle Boden-Dämmung covers mineral wool boards used in floor and roof applications — correct product category. Density and λ-value within expected range for flat roof insulation.
Waterproofing membrane	MEDIUM	A bitumen cap sheet EPD was not found in this datastock under "Dachdichtungsbahn". PE-HD membrane is used as a proxy — it serves the same waterproofing function but has different environmental characteristics. Flag for replacement if a bitumen EPD is found.
Gravel ballast	HIGH	Kies 2/32 is the standard grading for flat roof ballast per DIN 18531. Direct name and application match. Typical density 1,700–1,800 kg/m³.

SECTION 06

Python Code

Run in Google Colab or any Python 3.10+ environment. No API key required — the Ökobaumat API is public. Full script available as *Q2_simple_roof_matcher.py*.

```
# Install if needed: pip install requests import requests DATASTOCK = "cd2bda71-760b-4fcc-8a0b-3877c10000a8" # OEKOBAUDAT 2021-II BASE_URL = f"https://oekobaudat.de/OEKOBAU.DAT/resource/datastocks/{DATASTOCK}/processes" layers = [ {"layer": "Structural slab", "query": "Beton", "thickness_mm": 120}, {"layer": "Vapour barrier", "query": "Bitumenbahn", "thickness_mm": 5}, {"layer": "Thermal insulation", "query": "Mineralwolle", "thickness_mm": 60}, {"layer": "Waterproofing", "query": "Abdichtung", "thickness_mm": 5}, {"layer": "Gravel ballast", "query": "Kies", "thickness_mm": 10}, ] for item in layers: params = {"search": "true", "name": item["query"], "format": "json", "pageSize": 3} r = requests.get(BASE_URL, params=params, timeout=10) data = r.json().get("data", []) print(f"\n=== {item['layer']} ({item['thickness_mm']} mm) ===") for p in data: raw = p.get("name") name = raw[0].get("value", "") if isinstance(raw, list) else raw print(f" {p.get('uuid')} {name}")
```

SECTION 07

Assumptions & Limitations

- Roof type interpreted as a warm flat roof (Warmdach / nicht belüftetes Flachdach) — the most common generic flat roof construction in Germany
- Layer order: inside (structural) → outside (weather-exposed), per standard flat roof build-up
- Insulation thickness (60 mm) set to fill the 200 mm budget after structural slab, membranes and ballast — not GEG-compliant; a real design would use 120–160 mm
- Datastock fixed to OEKOBAUDAT 2021-II — newer datastocks (e.g. OBD_2024_I_A2) may have different EPD availability
- Waterproofing match is a PE membrane proxy — a dedicated bitumen cap sheet EPD was not found in this datastock under tested keywords
- Reinforcement steel for the concrete slab is not separately modelled — no combined Stahlbeton EPD available in this datastock
- EPD search is keyword-based — the Ökobaudat API does not support semantic search; match quality depends on keyword alignment
- No LCA values calculated in this approach — EPD matching only. GWP / PERT / PENRT would require quantity × EPD declared value calculation (see Q1 for methodology)

Q2 — EPD Matching · Generic Roof 200mm ·
OEKOBAUDAT 2021-II

Simple approach · Claude LLM + REST API · DIN 18531 · DIN
4108-2 · DIN 1045